

Summary

Results-oriented Python Developer with a background in data processing. Capable of developing robust solutions using Python libraries and ensuring seamless data flow. Competent in API testing with Postman and automating deployment processes using CI/CD tools like GitHub Actions. ready to leverage strong analytical skills to build high-quality software solutions.

Skills

Programming:

Python 3

Core Concepts

OOP, Data Structures, Algorithms, RESTful APIs, Error Handling, File handling, SDLC.

Frameworks & Libraries

NumPy, Pandas, SciPy, Matplotlib, Keras, PyTorch, TensorFlow, Jupyter

Automation

ETL Basics, Selenium, BeautifulSoup

Machine Learning:

Regression, Classification, Clustering, CNN, LLM, NLP, Model Optimization

Database

MySQL, PostgreSQL, SQL Server, Query optimization

Data Handling & Analysis

Data preprocessing, Data Validation, Data Quality & Governance, Exploratory Data Analysis (EDA).

Web frameworks

Flask, Streamlit

Big Data & Cloud:

Hadoop, Spark, AWS

Tools & Platforms

Git/GitHub(Version Control), Docker

Experience

infotact

Data Science and ML Intern

Dec 2025 - Present

Bangalore

Publications

Machine Learning Approaches in Drug Classification from biomarkers of Epileptic Seizure

Recent innovation in science , communication, computation

April 2024

Identification of Autism Spectrum Disorder (ASD) using face detection from images of toddlers and teens

Co-authored research on ASD detection using face recognition techniques.

April, 2025

Projects Experience

Lung Cancer Detection Using CNN

Built a multi-class lung cancer detection model using CNNs trained on 4,000+ CT-scan images covering Adenocarcinoma, Large Cell Carcinoma, Squamous Cell Carcinoma, and Normal cases.

jan 2024

Designed a custom CNN achieving **89.7%** accuracy, later improved to **94.3%** accuracy and **93.1%** recall using **InceptionV3**, significantly reducing false negatives crucial for early diagnosis.

Improved generalization by **+12%** through normalization, denoising, and augmentation while reducing overfitting by **15%**.

Conducted model comparison where InceptionV3 achieved **95%** precision, outperforming the baseline CNN by **18%** in classification performance.

Optimized batching and data pipelines, lowering training time per epoch by **32%** for faster experimentation.

FactoryGuard AI: IoT Predictive Maintenance Engine

Developed an end-to-end Machine Learning pipeline to predict equipment failure **24 hours in advance** using NASA CMAPSS sensor data, aiming to reduce unplanned downtime costs.

Engineered time-series features and handled **imbalanced data** (<5% failures) using **SMOTE**, achieving high recall for critical failures.

Trained and optimized an **XGBoost Classifier**, utilizing **SHAP** values to provide interpretable "Red Light" warnings for specific sensor malfunctions.

Deployed the model as a real-time REST API using **Flask** and containerized the application with **Docker** for production-ready scalability.

Predictive Modeling for Customer Tipping Behavior in NYC Taxis

Analyzed the NYC TLC dataset containing 50,000+ trip records to model customer tipping behavior.

July 2025

Engineered trip-level features (duration, fare ratios, temporal patterns), improving model accuracy by **+14%**.

Developed classification models including Logistic Regression, Random Forest, and XGBoost, achieving **92%** accuracy and **0.91** F1-score with XGBoost.

Improved precision by **+11%** through noise reduction, outlier handling, and categorical encoding.

Specialization

Google Advanced Data Analytics

 [Certification](#)

Completed specialization covering foundations of Data Science, Python, Statistics, Regression Analysis, Machine Learning, Data Analytics.

Nov 2025

IBM Machine Learning

 [Certification](#)

Completed 6-course ML specialization covering supervised, unsupervised learning, and time-series methods.

Sep 2025

Education

JIS UNIVERSITY

Kolkata

8.25 CGPA

2021 - 2025

B.Tech on Computer Science and Engineering

Languages

English

Hindi

Bangali

Odia